

A KIRKMAN TRIPLE SYSTEM OF ORDER 27 WITH A STEINER TRIPLE SYSTEM OF ORDER 7 AS A SUBDESIGN

ABSTRACT. Dukes and Lamken write that, although a $\text{KTS}(27)$ with a sub- $\text{STS}(u)$ is easy to construct for $u = 3, 9, 13$, “it appears difficult to find a $\text{KTS}(27)$ with an $\text{STS}(7)$ subdesign”. We exhibit one. Its thirteen parallel classes are printed in full, and everything claimed about them is elementary counting on the printed table. Whether such a design has been exhibited before was not determined here; no search of the literature or of design databases was carried out. The orders $v = 33$ and $v = 39$ of the same sentence of Dukes and Lamken are untouched.

1. THE PROBLEM

A *Steiner triple system* $\text{STS}(v)$ is a set of v points together with a collection of 3-subsets, called blocks or triples, covering every pair of points exactly once. It is a *Kirkman triple system* $\text{KTS}(v)$ if its blocks can be partitioned into *parallel classes*, each a partition of the point set; there are then necessarily $(v - 1)/2$ of them. A *subdesign* of order u is a u -subset W of the points together with the blocks lying inside W , this being required to be an $\text{STS}(u)$.

Dukes and Lamken [3] write:

There remain a number of interesting and difficult computational problems in finding KTS with STS subdesigns. We can construct $\text{KTS}(v)$ with all possible admissible $\text{STS}(u)$ for the two smallest cases, $v = 15$ and $v = 21$. But, although it is easy to construct $\text{KTS}(27)$ with sub- $\text{STS}(u)$ for $u = 3, 9, 13$, it appears difficult to find a $\text{KTS}(27)$ with an $\text{STS}(7)$ subdesign. Similarly, the existence of $\text{KTS}(v)$ with an $\text{STS}(7)$ subdesign is unknown for $v = 33$ and 39. New direct constructions are needed for these cases.

The passage is quoted from the e-print arXiv:2110.07874v1 of [3].

Two conventions of [3] decide what counts as a witness. First, the subdesign notion is the one recalled above: a point subset together with the blocks inside it. Second, no compatibility with the resolution is imposed — [3] writes that “a ‘Kirkman subsystem’ or sub- $\text{KTS}(u)$ is a subdesign which is resolvable in such a way that parallel classes of the $\text{KTS}(u)$ are restrictions of parallel classes of the $\text{KTS}(v)$ ”. We do not assume such extra structure

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for subdesigns here.” At $u = 7$ the stronger notion cannot apply in any case, since $3 \nmid 7$ and a set of 7 points has no partition into triples.

The problem is attributed in [3] to Open Problem 4 of Stinson’s survey [5], with [2, §19.7] named there as the section recording its status.

2. THE DESIGN

Let $V = \{0, 1, \dots, 26\}$ and $U = \{0, 1, 2, 3, 4, 5, 6\}$. Table 1 lists thirteen sets $C_0, \dots, C_{12}$ of nine triples each, each set written on two lines, the first ending in a vertical bar to mark the continuation. Call the resulting collection of $13 \cdot 9 = 117$ triples $\mathcal{D}$.

TABLE 1. The thirteen parallel classes. Points are $0, \dots, 26$; triples within a class are separated by |, and a trailing | means the class continues on the next line.

C00:	0 1 3 2 17 18 4 11 21 5 22 23 6 12 19
	7 9 14 8 15 20 10 13 24 16 25 26
C01:	0 7 8 1 2 4 3 20 24 5 19 26 6 18 25
	9 10 11 12 15 22 13 16 23 14 17 21
C02:	0 11 16 1 19 25 2 3 5 4 9 20 6 7 24
	8 21 26 10 14 22 12 13 18 15 17 23
C03:	0 9 17 1 10 18 2 20 26 3 4 6 5 7 25
	8 22 24 11 12 23 13 14 19 15 16 21
C04:	0 4 5 1 16 20 2 10 23 3 14 18 6 21 22
	7 11 13 8 17 19 9 12 26 15 24 25
C05:	0 10 15 1 5 6 2 11 19 3 7 26 4 18 24
	8 23 25 9 13 21 12 14 20 16 17 22
C06:	0 2 6 1 9 22 3 21 23 4 15 19 5 13 20
	7 10 12 8 16 18 11 14 25 17 24 26
C07:	0 19 23 1 15 26 2 7 22 3 13 17 4 8 12
	5 14 24 6 9 16 10 21 25 11 18 20
C08:	0 20 21 1 8 13 2 16 24 3 12 25 4 7 23
	5 10 17 6 14 15 9 18 19 11 22 26
C09:	0 18 22 1 7 21 2 8 14 3 11 15 4 17 25
	5 12 16 6 13 26 9 23 24 10 19 20
C10:	0 14 26 1 12 17 2 9 25 3 10 16 4 13 22
	5 8 11 6 20 23 7 15 18 19 21 24
C11:	0 12 24 1 14 23 2 13 15 3 8 9 4 10 26
	5 18 21 6 11 17 7 16 19 20 22 25
C12:	0 13 25 1 11 24 2 12 21 3 19 22 4 14 16
	5 9 15 6 8 10 7 17 20 18 23 26

Theorem 1. $\mathcal{D}$ is a Kirkman triple system of order 27 with $C_0, \dots, C_{12}$ as its parallel classes, and U carries an STS(7) subdesign of $\mathcal{D}$. In particular a KTS(27) with an STS(7) subdesign exists.

Proof. Each of the thirteen lists in Table 1 contains nine triples whose 27 entries are $0, 1, \dots, 26$ without repetition, so each C_{ij} is a partition of V and $\mathcal{D}$ is the disjoint union of thirteen parallel classes. The 117 triples carry $117 \cdot 3 = 351 = \binom{27}{2}$ pairs, and the pairs so obtained are all distinct; being $\binom{27}{2}$ in number they are therefore all of them, each once. Hence $\mathcal{D}$ is an STS(27) and $C_{00}, \dots, C_{12}$ is a resolution of it, i.e. a KTS(27); the number of classes is $(27 - 1)/2 = 13$.

The triples of $\mathcal{D}$ contained in U are exactly

$$\{0, 1, 3\}, \{1, 2, 4\}, \{2, 3, 5\}, \{3, 4, 6\}, \{0, 4, 5\}, \{1, 5, 6\}, \{0, 2, 6\},$$

one in each of $C_{00}, C_{01}, \dots, C_{06}$ in that order. They carry $7 \cdot 3 = 21 = \binom{7}{2}$ pairs, all distinct, hence all pairs inside U , each once. So U together with these seven triples is an STS(7): it is the Fano plane, being the unique STS(7). By the definition recalled in §1 it is a subdesign of $\mathcal{D}$ of order 7. $\square$

Remark. No triple of $\mathcal{D}$ meets U in exactly two points: such a pair lies inside U and is already covered by one of the seven triples above, and pairs are covered once. So the 117 triples split as $7 + 70 + 40$: seven inside U , seventy meeting U in one point, and forty disjoint from U . Concretely, each of $C_{00}, \dots, C_{06}$ consists of one inner, four one-point and four disjoint triples, and each of $C_{07}, \dots, C_{12}$ of seven one-point and two disjoint triples.

3. THE SUBDESIGNS OF THE PRINTED DESIGN

Write $\langle S \rangle$ for the *closure* of a set S of points: the smallest subset containing S and closed under the operation sending a pair of its points to the third point of the block through them. A closed set of size at least 3, with the blocks inside it, is a subdesign.

Lemma 2. *If W is a subdesign of order $w \geq 3$ of an STS(v) and W is proper, then $v \geq 2w + 1$.*

Proof. Fix $x \notin W$. For $p \in W$ let $f(p)$ be the third point of the block through x and p . Then $f(p) \notin W$, since a block meeting W in two points has its third point in W and would give a pair of W covered twice; and $f(p) \neq x$. If $f(p) = f(q)$ with $p \neq q$ then the block $\{x, p, f(p)\}$ contains q , so $p, q, f(p)$ are collinear and $f(p) \in W$, a contradiction. Hence f is injective into $V \setminus (W \cup \{x\})$, and $v \geq w + 1 + w$. $\square$

Proposition 3. *$\mathcal{D}$ has exactly one subdesign of order 7, namely U , and none of order 9 or 13. Consequently the number of order-7 subdesigns through a point equals 1 for each of the seven points of U and 0 for each of the other twenty; hence $\text{Aut}(\mathcal{D})$ is not transitive on points.*

Proof. Let W be a subdesign of $\mathcal{D}$ of order $u \in \{7, 9, 13\}$ and let $x, y, z \in W$ be three points not on a common block; such a triple exists because $u \geq 7$. Then $\langle x, y, z \rangle \subseteq W$ is a subdesign of some order u' with $3 < u' \leq u$, and

u' is an admissible order for a Steiner triple system, hence $u' \in \{7, 9, 13\}$ with $u' \leq u$. If $u' < u$ then Lemma 2 forces $u \geq 2u' + 1 \geq 15$, which is false. So $\langle x, y, z \rangle = W$: every subdesign of order 7, 9 or 13 is the closure of a triple of its own points. Closing each of the $\binom{27}{3} = 2925$ triples of points of $\mathcal{D}$ therefore produces every such subdesign, and doing so yields exactly one closed set of size 7, namely U , and none of size 9 or 13. (The closed sets of size 3 are the 117 blocks.)

The set of points lying on a subdesign of order 7 is preserved by every automorphism, and so is the number of such subdesigns through a given point. That number takes two values, 1 and 0, so $\text{Aut}(\mathcal{D})$ has more than one orbit on V . $\square$

4. A COMPACT FORM

The design is invariant under an automorphism of order 3, which allows it to be recorded by five of the thirteen classes. Let

$$\sigma = (1\ 2\ 4)(3\ 6\ 5)(9\ 10\ 11)(12\ 13\ 14)(15\ 16\ 17)(18\ 19\ 20)(21\ 22\ 23)(24\ 25\ 26),$$

fixing 0, 7, 8 and no other point. Then σ has order 3, it permutes the 117 triples of $\mathcal{D}$ among themselves, and on U it acts as the collineation $x \mapsto 2x \bmod 7$ of the Fano plane. Its fixed set $\{0, 7, 8\}$ is itself a block; that block lies in C_{01} . On classes, σ induces

$$\pi = (C_{00}\ C_{06}\ C_{04})(C_{02}\ C_{03}\ C_{05})(C_{07}\ C_{08}\ C_{09})(C_{10}\ C_{11}\ C_{12}),$$

fixing C_{01} . The five classes $C_{00}, C_{01}, C_{02}, C_{07}, C_{10}$ meet every π -orbit, so applying σ and σ^2 to them and placing the images in the classes named by π recovers all thirteen classes of Table 1. Five base classes therefore suffice to record the design; Table 1 remains the definition.

5. SCOPE

A KTS(27) with an STS(7) subdesign exists; one is printed in Table 1. In the terms of [3] this is the cell $(v, u) = (27, 7)$; the orders $v = 33$ and $v = 39$ of the same sentence are untouched, and Open Problem 4 of [5] is not settled here. No priority is claimed, and no novelty either: whether a KTS(27) with an STS(7) subdesign has been exhibited before, in [4], in [2, §19.7], in the published version of [3], or in a design database, was not checked here, so this note does not establish that any still-open request is answered by it. Nothing is claimed about the number of such designs, about isomorphism (none was rejected), or about the order of $\text{Aut}(\mathcal{D})$ beyond its containing σ and, by Proposition 3, not being transitive on points.

6. VERIFICATION

Both parts of Theorem 1 are inspections of Table 1, and no input other than that table is used anywhere above.

A verification program `verify.py` accompanies this paper, together with the transcript of a run. It parses Table 1 from the printed text, in exact

integer arithmetic and using only the Python standard library, and re-derives every quantity stated here: the resolution and the 351 pairs, the seven inner blocks and their Fano structure, the $7 + 70 + 40$ census, the order and action of σ and the induced π , the reconstruction from the five base classes, and the subdesign census of Proposition 3 by closing all 2925 point triples. All 59 checks pass. The transcript also mentions a second design found in the underlying search; it is not printed here and nothing about it is claimed or verified.

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